

Problem Set for 07 April 2025

Exercise Let V be a vector space over $\mathbb{F}$, and define $\text{End}(V) := \text{Hom}_{\mathbb{F}}(V, V)$. Define:

- $f \in \text{End}(V)$ as *almost zero* (denoted $f \in \text{AZ}(V)$), provided $\dim \text{im}(f) < \infty$.
- $f \in \text{End}(V)$ as *almost an isomorphism* (denoted $f \in \text{AI}(V)$), provided $\dim \ker f + \dim \frac{V}{\text{im}(f)} < \infty$.

Now finish the following.

1. If $\dim V < \infty$, determine $\text{AZ}(V)$ and $\text{AI}(V)$.

$$\text{AZ}(V) = \text{AI}(V) = \text{End}(V).$$

2. If $V = \mathbb{R}[x]$, find an injective $f \in \text{AI}(V)$ and a surjective $g \in \text{AI}(V)$. Additionally, find some $h \notin \text{AI}(V) \cup \text{AZ}(V)$.

f 右移, g 左移. 将 $\mathbb{R}[x]$ 依照奇偶函数写作 $V_1 \oplus V_2$, 则

$$\varphi : V_1 \oplus V_2 \rightarrow V_1 \oplus V_2, \quad (f, g) \mapsto (f, 0)$$

满足 $\dim \ker \varphi, \dim \text{im } \varphi, \dim \frac{\mathbb{R}[x]}{\text{im } \varphi} = \infty$.

3. Determine whether $\text{AZ}(V)$ is a subspace of $\text{End}(V)$. What about $\text{AI}(V)$?

先看 $\text{AZ}(V)$, 下分别证明 $\text{AZ}(V)$ 对数乘和加法封闭.

1. $f \in \text{AZ}(V)$, 则 $\text{im}(\lambda f) \subset \text{im}(f)$ 是有限维的.
 2. $f, g \in \text{AZ}(V)$, 则 $\text{im}(f + g) \subset \text{im}(f) + \text{im}(g)$ 是有限维的.
- $0 \notin \text{AI}(V)$, 从而 $\text{AI}(V)$ 不是线性子空间.

4. Show that for arbitrary $f, g \in \text{AZ}(V)$, one has $g \circ f \in \text{AZ}(V)$.

下一问的弱化.

5. Prove that $\text{AZ}(V)$ is an **ideal** in $\text{End}(V)$. The term **ideal** is defined in 20 March 2025.

若 $f \in \text{AZ}(V)$, 由满射

$$\varphi : \text{im}(f) \twoheadrightarrow \text{im}(g \circ f), \quad f(v) \mapsto g(f(v)),$$

此时 $\dim \operatorname{im}(g \circ f) \leq \operatorname{im}(f)$. 从而 $g \circ f \in \operatorname{AZ}(V)$.

若 $g \in \operatorname{AZ}(V)$, 显然 $\operatorname{im}(g \circ f) \subset \operatorname{im}(g)$ 是有限维的. 从而 $g \circ f \in \operatorname{AZ}(V)$.

$\operatorname{AZ}(V)$ 是线性空间, 且

$$\operatorname{AZ}(V) \circ \operatorname{End}(V) \subset \operatorname{AZ}(V) \supset \operatorname{End}(V) \circ \operatorname{AZ}(V).$$

从而是理想.

6. Show that for arbitrary $f, g \in \operatorname{AI}(V)$, one has $g \circ f \in \operatorname{AI}(V)$.

只需证明以下事实.

1. $\dim \ker f + \dim \ker g \geq \dim \ker(g \circ f)$.

2. $\dim \frac{V}{\operatorname{im}(f)} + \dim \frac{V}{\operatorname{im}(g)} \geq \dim \frac{V}{\operatorname{im}(g \circ f)}$.

第一问, 构造

$$f_s : \ker(g \circ f) \rightarrow \ker(g), \quad x \mapsto f(x).$$

这一映射的 image 是 $\ker g$ 的子空间, kernel 是 $\ker(g \circ f) \cap \ker f = \ker f$. 从而

$$\dim \ker(g \circ f) = \dim \ker f + \dim \ker g - \dim \frac{\ker g}{f(\ker(g \circ f))}.$$

由 $\dim \frac{\ker g}{f(\ker(g \circ f))} \leq \dim \ker g < \infty$, 左式是有限的.

第二问, 构造

$$g_q : \frac{V}{\operatorname{im}(f)} \rightarrow \frac{V}{\operatorname{im}(g \circ f)}, \quad (v + f(V)) \mapsto (g(v) + g(f(V))).$$

这一 g_q 是良定义的, 因为

$$\Phi : V \rightarrow \frac{V}{\operatorname{im}(g \circ f)}, \quad v \mapsto (g(v) + g(V))$$

的 kernel 包含了 $\operatorname{im}(f)$, 从而 Φ 可以定义在商空间上, 得 g_q . 特别地,

$$\begin{aligned}
\frac{V/\text{im}(g \circ f)}{\text{im}(g_q)} &= \frac{V/\text{im}(g \circ f)}{\text{im}(\Phi)} \quad \text{由 } \Phi = g_q \circ \text{“某满射”} \\
&= \frac{V/\text{im}(g \circ f)}{\{g(v) + \text{im}(g \circ f)\}} \quad \text{由定义} \\
&= \frac{V/\text{im}(g \circ f)}{\text{im}(g)/\text{im}(g \circ f)} \quad \text{写作商空间的子空间} \\
&= \frac{V}{\text{im}(g)} \quad \text{同构定理.}
\end{aligned}$$

因此 $g_q(\frac{V}{\text{im}(f)})$ 是满射 $\frac{V}{\text{im}(g \circ f)} \twoheadrightarrow \frac{V}{\text{im}(g)}$ 的 kernel. 从而

$$\dim \frac{V}{\text{im}(g \circ f)} = \dim \frac{V}{\text{im}(g)} + \dim \frac{V}{\text{im}(f)} - \dim \ker g_q.$$

由 $\dim \ker g_q \leq \dim \frac{V}{\text{im}(f)} < \infty$, 左式是有限的.

7. Demonstrate that if any two elements of $\{f, g, g \circ f\}$ belong to $\text{AI}(V)$, then so does the third.

Hint: for $f \in \text{AI}$, set $\Phi(f) := \dim \ker f - \dim \frac{V}{\text{im}(f)}$, show that $\Phi(f \circ g) = \Phi(f) + \Phi(g)$.

Hint 中的等式成立, 当且仅当上述

$$\dim \frac{\ker g}{f(\ker(g \circ f))} = \dim \ker g_q.$$

实际上,

$$\begin{aligned}
1. \ker g_q &= \frac{\ker g + \text{im } f}{\text{im } f}; \\
2. \frac{\ker g}{f(\ker(g \circ f))} &= \frac{\ker g}{\text{im } f \cap \ker g}.
\end{aligned}$$

得证.

8. Show that for arbitrary $f \in \text{AZ}(V)$, one has $(\text{id}_V + f) \in \text{AI}(V)$.

若 $(\text{id}_V + f)(v) = 0$, 则 $v \in \text{im } f$. 从而 $\ker(\text{id}_V + f) \subset \text{im } f$ 是有限维的.

对偶地, 只需证明存在满射 $\text{im } f \twoheadrightarrow \frac{V}{\text{im}(\text{id}_V + f)}$, 依照同构定理, 即证 $\ker f \subset \text{im}(\text{id}_V + f)$. 若 $f(v) = 0$, 则 $v = (\text{id}_V + f)(v)$, 得证.

9. **(Optional)** By Axiom of Choice, every subspace has a complementary subspace. Prove that $f \in \mathbf{AI}(V)$ can always be expressed as the summation of a map of $\mathbf{AZ}(V)$ -type and an automorphism (i.e., auto = iso + endo). Conversely, show that any such summation belongs to $\mathbf{AI}(V)$.

对 f , 考虑

$$V = \ker f \oplus V_1 \xrightarrow{f} \operatorname{im}(f) \oplus V_2 = V.$$

以上诱导的 $V_1 \rightarrow \operatorname{im} f$ 是同构. 由于 $\ker f$ 和 V_2 是有限维的, 任何通过 $\ker f$ 或 V_2 分解的映射都是 $\mathbf{AZ}(V)$ -类型的. 可以将 f 写作列向量之间的矩阵, 即

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} : \ker f \oplus V_1 \rightarrow \operatorname{im}(f) \oplus V_2, \quad (x, y) \mapsto (a(x) + b(y), c(x) + d(y)).$$

此时 $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} a & 0 \\ c & d \end{pmatrix}$ 就是所求的分解.

10. **(Optional)** Show that $\frac{\operatorname{End}(V)}{\mathbf{AZ}(V)}$ is a quotient ring with a well-defined subset $\frac{\mathbf{AI}(V)}{\mathbf{AZ}(V)}$, consisting precisely of the invertible elements. In this quotient ring, the operators $\frac{d}{dx}(-)$ and $\int_0^x(-) dx$ are mutually inverse.

本小问面向熟悉环论的同学, 其实没什么新东西.

Problem (Optional) Let V be a vector space over $\mathbb{F}$. We define

$$\prod_{i \in I} V := \underbrace{V \times \cdots \times V}_{I\text{-copies}} \simeq \text{Hom}_{\text{Sets}}(I, V)$$

as an $\mathbb{F}$ -vector space whose elements are sequences $(v_i)_{i \in I}$. Informally, $\prod_{i \in I} V$ represents the space of sequences indexed by I with values in V .

1. Define $e_i := (\dots, 0, 0, \underbrace{1_{\mathbb{F}}}_{i\text{-th entry}}, 0, 0, \dots) \in \prod_{i \in I} \mathbb{F}$. Determine when $\{e_i\}_{i \in I}$ forms a basis of $\prod_{i \in I} \mathbb{F}$.

当且仅当 I 有限.

2. Describe $\text{span}(\{e_i\}_{i \in I})$ in natural language.

$\prod_{i \in I} \mathbb{F}$ 中有且仅有有限项非零的东西.

3. Substituting I with $V_{\text{as a set}}$, the underlying set of V as an $\mathbb{F}$ -vector space, prove that π is a linear surjection and determine the equivalence relation $R_{\mathbb{F}}$ as defined in 24 March 2025:

$$\pi : \text{span}(\{e_v\}_{v \in V_{\text{as a set}}}) \rightarrow V, \quad e_v \mapsto v.$$

映射层面, $\text{id}_{V_{\text{as a set}}} : \{e_v\}_{v \in V_{\text{as a set}}} \rightarrow V$ 是恒等的. 由于 $\{e_v\}_{v \in V_{\text{as a set}}}$ 是 $\prod_V \mathbb{F}$ 中的线性无关组, 自然存在唯一定义的

$$\pi : \text{span}(\{e_v\}_{v \in V_{\text{as a set}}}) \rightarrow V, \quad e_v \mapsto v.$$

使得限制在基上的映射为 $\pi|_{\{e_v\}_{v \in V_{\text{as a set}}}} = \text{id}_{V_{\text{as a set}}}$.

这时候, $R_{\mathbb{F}}$ 就是 V 中本有的线性组合关系.

Remark 以上问题就是将线性空间的定义重新表述了一遍: 结构 = 自由定义的结构商去关系.